CHENYANG LAI

chenyang.lai@ntu.edu.sg | | | Google Scholar | Homepage | Scopus ID: 57210472624

Research Fellow, Department of Mechanical and Aerospace Engineering
Nanyang Technological University

RESEARCH INTERESTS

- Physics-informed Machine Learning (PIML)
- Industrial AI
- AI Safety and Trustworthiness
- Manufacturing Process Modeling and Monitoring
- Prognostics and Health Management (PHM)

EMPLOYMENT

Research Fellow 05/2026 – Present
Nanyang Technological University (NTU), Singapore
Mentor: Dr. Xuan Liang

- Thermal and deformation-field prediction for advanced additive manufacturing using PIML methods.

Postdoctoral Researcher 02/2025 – 05/2026
Politecnico di Milano (POLIMI), Milan, Italy
Mentors: Prof. Piero Baraldi and Prof. Enrico Zio

- Condition monitoring and degradation assessment of power electronics used in wind turbines.
- Anomaly detection of power transistors using scattering parameters.
- Reinforcement learning for limp-mode management of power converters.

EDUCATION

Politecnico di Milano (POLIMI) Milan, Italy
Ph.D. in Energy and Nuclear Science and Technology, *Cum Laude* 01/2021 – 07/2025
Supervisors: Prof. Piero Baraldi and Prof. Enrico Zio

- Thesis: Physics-informed deep learning methods of prognostics and health management for predictive maintenance of industrial equipment

Northwestern Polytechnical University (NPU) Xi'an, China
Master of Control Science and Engineering, *Cum Laude* 08/2017 – 04/2020

Harbin Engineering University (HEU) Harbin, China
Bachelor of Science in Energy and Power Engineering 08/2013 – 07/2017

RESEARCH PROJECTS

Ministry of Education (MOE) Project, Singapore 05/2026 – Present
Physics-Enhanced Machine Learning-based Simulation with Applications to Structural Design and Process Optimization for Metal Additive Manufacturing

- Transfer-learning-enhanced physics-informed neural network for unsupervised full-field distortion prediction of metal additive manufacturing by directed energy deposition.
- Physics-informed neural network for thermal-field prediction of the directed energy deposition process.

European Union Project: SAFEPOWER 01/2025 – 06/2026
Safer and More Reliable WBG/UWBG-Based MVDC Power Converters

- AI methods for degradation assessment and remaining useful life prediction of IGBTs in wind turbine converters, coordinating with the University of Warwick, UK.
- Convolutional neural network to predict the degradation evolution of IGBT devices from S-parameters, coordinating with the University of Technology Tarbes Occitanie Pyrenees (UTTOP), France.
- Reinforcement learning framework for limp-mode management of power converters, coordinating with Power Electronics España, S.L.

European Union Project: MOSAIC 04/2025 – 06/2026
A Mosaic of Essential Electronic Components and Systems for Our Automated Digital Future in Industry and Mobility

- Zero-shot anomaly detection for semiconductor manufacturing based on contrastive language–image pre-training (CLIP), coordinating with Infineon Technologies Austria AG.

European Union Project: iRel40 01/2021 – 04/2023
Intelligent Reliability 4.0

- Feature selection and AI methods for case-temperature estimation of IGBTs used in induction cooktops, coordinating with BSH Electrodomésticos España, Spain.
- Physics-informed domain adaptation and AI methods for health assessment and remaining useful life prediction of electronic components, coordinating with BSH Electrodomésticos España, Spain.

JOURNAL PUBLICATIONS

12. **Lai, C.**, & Liang, X.*. (2026). Transfer learning-enhanced PINN for unsupervised deformation prediction of directed energy deposition. In preparation.
11. **Lai, C.**, Baraldi, P.*, & Zio, E. (2026). PIDEL: Physics-Informed Deep Evidential Long Short-Term Memory Networks for Remaining Useful Life Prediction with Uncertainty Quantification of Electronic Components. In preparation.
10. **Lai, C.***, Liang, X., Ran, L., & Zio, E. (2026). Few-Shot Anomaly Detection under Data Scarcity Using Pretrained Vision-Language Representations and Dropout-Calibrated Decision Boundaries. *Advanced Engineering Informatics*. Under review.
9. **Lai, C.**, Baraldi, P.*, Baldo, L., Bertone, M., Dalla Vedova, M. D. L., Quattrocchi, G., & Zio, E. (2026). Physics-informed long short-term memory networks with transfer learning for anomaly detection in electromechanical actuators. *Engineering Applications of Artificial Intelligence*. Under review.
8. Liu, A., Qu, C., **Lai, C.**, Zhu, H., & Wang, B.* (2026). SE-PLNN: A sample entropy and parallel liquid neural network framework for Remaining Useful Life prediction of mechanical systems under multi-fault degradation. *Measurement*, 292, 123459. (IF = 6.1)
7. Li, X., Lin, W.*, Wang, B., Shi, L., **Lai, C.**, Qiao, Q., et al. (2026). Comprehensive compound fault modelling and characterisation of building integrated photovoltaics based on thermal-electrical coupled simulation. *Solar Energy*, 315, 114790. (IF = 7.9)
6. **Lai, C.***, Baraldi, P.*, & Zio, E. (2026). Physics-informed domain adaptation for health state assessment of industrial components. *Reliability Engineering & System Safety*, 113021. (IF = 13.7)
5. Liu, H., Li, B.*, **Lai, C.**, & Zio, E. (2026). Unsupervised domain adaptation regression network with efficient channel attention mechanism for tool wear prediction under variable cutting conditions. *Journal of Manufacturing Processes*, 172, 1091–1105. (IF = 7.8)
4. **Lai, C.***, Wang, B.*, Fan, S., & Zio, E. (2026). Self-balancing physics-informed LSTM for soft-sensing of slurry concentration in dredger operation. *Advanced Engineering Informatics*, 69, 103978. (IF = 11.5)
3. **Lai, C.**, Baraldi, P.*, & Zio, E. (2025). Gradient-enhanced physics-informed long short-term memory networks for stable and accurate prediction of the RUL of electronic components. *Reliability Engineering & System Safety*, 111485. (IF = 13.7)
2. **Lai, C.**, Baraldi, P.*, & Zio, E. (2024). Physics-informed deep autoencoder for fault detection in new-design systems. *Mechanical Systems and Signal Processing*, 215, 111420. (IF = 10.2)
1. **Lai, C.**, Ahmed, I.*, Zio, E.*, Li, W., Zhang, Y., Yao, W., & Chen, J. (2024). A multistage physics-informed neural network for fault detection in regulating valves of nuclear power plants. *Energies*, 17(11), 2647. (IF = 3.9)

CONFERENCE AND WORKSHOPS

15. **Lai, C.**, Baraldi, P.*, Moyo, W., Baffreau, S., Liang, X.*, Vidal, P., Vine, G., & Zio, E. (2026). Unsupervised Degradation Evolution Tracking of Power Transistors Using S-Parameters based on Convolutional Neural Network Autoencoders. *Procedia CIRP* (CIRPe 2026, Nanyang Technological University, Singapore). (Proceeding & Presentation)
14. Liu, H., Shen, M., **Lai, C.***, & Liu, Z. (2026). PIASANet: Physics-informed adaptive spectral attention network for cross-load bearing fault diagnosis. *Proceedings of the 10th International Conference on Reliability Engineering (ICRE)*. (Proceeding & Presentation)
13. **Lai, C.**, Baraldi, P.*, Zinchenko, O., Zhang, X., Ran, L., & Zio, E. (2026). Assessment of the degradation of IGBTs used in wind turbine power converters by a method based on physics-informed transformers. *Proceedings of the 36th European Safety and Reliability Conference (ESREL 2026)*, 3113–3120. (Proceeding & Presentation)
12. **Lai, C.**, Baraldi, P.*, Aruna, Baldo, L., Dalla Vedova, M. D. L., Quattrocchi, G., & Zio, E. (2025). Anomaly detection in power inverters of electromechanical actuators based on convolutional neural network and long short-term memory cells. *2025 9th International Conference on System Reliability and Safety (ICSRS)*, IEEE, 136–142. (Proceeding & Presentation)
11. **Lai, C.***, Baraldi, P., & Zio, E. (2025). AI strategies in C&HM for MVDC converters: Studies ongoing. *SAFEPOWER WP3 Technical Workshop*. (Presentation)
10. **Lai, C.**, Baraldi, P.*, Quattrocchi, G., Dalla Vedova, M. D. L., Baldo, L., Bertone, M., & Zio, E. (2024). Detection of abnormal conditions in electro-mechanical actuators by physics-informed long short-term memory networks. *PHM Society European Conference*, 8(1). (Proceeding & Presentation)
9. **Lai, C.**, Baraldi, P.*, Quattrocchi, G., Dalla Vedova, M. D. L., Baldo, L., Bertone, M., & Zio, E. (2023). Fault Detection in Electro-Mechanical Actuators by Physics-Informed Long Short-Term Memory Networks. *2023 7th International Conference on System Reliability and Safety (ICSRS)*, IEEE. (Abstract & Presentation)
8. **Lai, C.***, Baraldi, P., Ahmed, I.*, & Zio, E. (2022). Development of PHM Method for Power Electronic Devices Based on Physics-Informed Neural Networks. *iRel40 Use Case Workshop, Ljubljana, Slovenia.* (Poster & Presentation)
7. **Lai, C.**, Baraldi, P.*, Floreale, G., & Zio, E. (2022). Physics-Informed Long Short-Term Memory Network for the Prediction of the Remaining Useful Life (RUL) of Insulated-Gate Bipolar Transistors (IGBTs). *2022 6th International Conference on System Reliability and Safety (ICSRS)*, IEEE. (Abstract & Presentation)
6. Juybari, M. N., Ahmed, I.*, Baraldi, P., Del Cueto, A., Gil, J., **Lai, C.**, et al. (2022). Estimation of the case temperature of insulated gate bipolar transistors in induction cooktops by deep neural network. *2022 6th International Conference on System Reliability and Safety (ICSRS)*, IEEE, 259–263. (Proceeding & Presentation)
5. **Lai, C.**, Baraldi, P.*, Ahmed, I., Zio, E., Gil, J., & Llorente, S. (2022). Monitoring degradation of insulated gate bipolar transistors in induction cooktops by artificial neural networks. *Proceedings of the 32nd European Safety and Reliability Conference (ESREL 2022)*, 1338–1345. (Proceeding & Presentation)
4. **Lai, C.**, Baraldi, P.*, & Zio, E. (2022). A Physics-informed Autoencoder for Early Fault Detection of Electro-hydraulic Actuators. *2021 5th International Conference on System Reliability and Safety (ICSRS)*, IEEE. (Abstract & Presentation)
3. Mao, H., Guo, Y., Li, R., & **Lai, C.** (2019). Versatile simulation platform for turboshaft engine control system. *2019 Chinese Control Conference (CCC)*, IEEE, 7211–7216. (Proceeding & Presentation)
2. **Lai, C.**, Guo, Y., & Mao, H. (2019). Main fuel pump health monitoring method based on Kriging surrogate model. *2019 Chinese Control Conference (CCC)*, IEEE, 5112–5117. (Proceeding & Presentation)
1. Zhao, W., Guo, Y., & **Lai, C.** (2018). Sensor fault diagnosis and estimation based on multiple-model approach for aeroengine. *2018 IEEE CSAA Guidance, Navigation and Control Conference (CGNCC)*, 1–6. (Proceeding & Presentation)

AWARDS

- Best Presentation Award, Anomaly detection in power inverters of electromechanical actuators based on convolutional neural network and long short-term memory cells, 9th International Conference on System Reliability and Safety (ICSRS), IEEE, 2025.
- Best Presentation Award, Fault detection in electro-mechanical actuators by physics-informed long short-term memory networks, 7th International Conference on System Reliability and Safety (ICSRS), IEEE, 2023.
- Outstanding Postgraduate Award, Northwestern Polytechnical University, 2020.
- Second Prize in the National Postgraduate Mathematical Modeling Competition, 2018.

ACADEMIC ACTIVITIES

- Chair, session "T1-C: Energy, Sustainability & Remanufacturing", CIRPe 2026, Nanyang Technological University, Singapore, October 13–16, 2026.
- Chair, session "T2-B: Circular Manufacturing, Remanufacturing & Handling Systems", CIRPe 2026, Nanyang Technological University, Singapore, October 13–16, 2026.
- Invited speaker, Symposium on "AI-Driven Simulation, Design and Control for Manufacturing Processes", 2nd International Conference on Intelligent Manufacturing (ICIM), Singapore, July 27–31, 2026.
- Chair, special session "Physics-Informed Machine Learning for Prognostics and Health Management of Industrial Equipment", 10th International Conference on Reliability Engineering (ICRE), Hangzhou, China, July 19–21, 2026.
- Co-chair, 10th International Conference on Reliability Engineering (ICRE), Hangzhou, China, July 19–21, 2026.
- Chair, special session "SS-06: Physics-Informed Machine Learning for Reliability, Availability, Maintainability and Safety (RAMS)", European Safety and Reliability Conference (ESREL), Braga, Portugal, June 14–19, 2026.
- Chair, special session "SS-10: Development and Application of Methods for Enhancing the Reliability of Power Electronics Devices", European Safety and Reliability Conference (ESREL), Braga, Portugal, June 14–19, 2026.
- Invited scholar, Silk Road International Symposium (SRIS), Xi'an Jiaotong University, Xi'an, China, November 1–4, 2025.
- Journal Reviewer for:
  Mechanical Systems and Signal Processing
  Reliability Engineering & System Safety
  Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability
  IEEE Transactions on Industrial Informatics
  IEEE Transactions on Transportation Electrification
  IEEE Journal of Emerging and Selected Topics in Power Electronics
  Applied Soft Computing

TEACHING AND MENTORING

- Lecturer, continuing education course "RAM&PHM 4.0: Advanced Methods for Reliability, Availability, Maintainability, Prognostics and Health Management of Industrial Equipment", Politecnico di Milano, 2022, 2023, and 2025.
- Lecturer, "Artificial Intelligence and Advanced Simulation for the Safety, Reliability and Maintenance of Energy Systems", M.Sc. in Nuclear Engineering, Politecnico di Milano, 2022, 2023, and 2024.

Mentoring and supervision

- 2026–Present: Francesco Marco Riboli (M.Sc.) — AI-based degradation assessment for power electronic devices using scattering parameters.
- 2026–Present: Yunlu Bai (Ph.D.) — Physics-informed machine learning for high-altitude structural health monitoring.
- 2025–2026: Tibyan Gadalla Naeem Mustafa (M.Sc.) — Physics-informed neural networks for modelling high-burnup structure formation in nuclear fuel.
- 2023–2024: Aruna (M.Sc.) — Fault detection of power inverters in electromechanical actuators using a CNN-LSTM approach.
- 2023–2024: Camilla Battistini (M.Sc.) — Physics-informed neural networks for the safety analysis of nuclear reactors.

SKILLS

- **Languages:** Mandarin (native), English (working proficiency), Italian (limited working proficiency), Spanish (elementary).
- **Computing:** MATLAB/Simulink, Python, PyTorch, C/C++, Git, Amesim, Ansys, LTspice, LaTeX, Microsoft Office.
- **Soft skills:** self-learning, critical thinking, teamwork, problem solving.